

Kirtan Patel

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SKILLS

Languages: Python, JavaScript, HTML5, CSS3

Frameworks & Libraries: ReactJS, Node.js, Flask, Next.js

Databases: MongoDB, MySQL, PostgreSQL

Tools & Platforms: Git/GitHub, Docker, Postman, Vercel, Linux

Security: Applied cryptography (AES-256-GCM, X25519, HKDF, HMAC), vulnerability scanning (OSV.dev), LLM red-teaming & prompt-injection detection, tamper-evident audit logging, TCP/IP socket programming

WORK EXPERIENCE

Technology Solutions, Office of the Vice Chancellor for Innovation – UIC

Chicago, IL

IT Helpdesk Support Manager

Dec 2024 – Present

- Lead the IT helpdesk support team, owning tier-2 escalation in TeamDynamix (TDX) across classroom networking, AV systems, and campus-wide technology while coaching specialists through complex tickets to resolution.
- Serve on the hiring team for support specialists, interviewing and evaluating candidates for technical skill and the ability to support non-technical users, then onboarding new hires onto the team.
- Co-author and maintain the team's troubleshooting documentation, turning recurring issues into repeatable resolution steps that keep support consistent and shorten new-hire ramp-up.
- Diagnose and repair laptops and client devices, acting as the hands-on tier-2 resource for faculty, staff, and students across campus.
- Own printer and hardware operations for 30 units across East Campus end-to-end; streamlined the fleet to 24 active units and cut average downtime from 6 to 1, saving thousands annually.

Technology Solutions, Office of the Vice Chancellor for Innovation – UIC

Chicago, IL

Classroom Technology Support Specialist

Aug 2022 – Dec 2024

- Kept instruction running across East Campus classrooms and lecture halls, supporting AV systems, projectors, audio, and Echo360 lecture capture and resolving live issues quickly to protect class time.
- Deployed and configured computer labs for courses and campus events, handling software installs and access provisioning.
- Trained and mentored incoming specialists to independent troubleshooting through shadowing, hands-on practice, and feedback on real tickets.

OPEN SOURCE

llm-thorn | Python, FastAPI, httpx, Pydantic, SQLite, PyPI

Author & Maintainer

- Authored and published (PyPI: **llm-thorn**) a runtime semantic security layer that sits as a reverse proxy in front of any LLM, screening every request and response through layered detectors – heuristic, semantic intent, multi-turn context risk, and output analysis – governed by a YAML policy-as-code engine with five enforcement actions (allow, warn, block, redact, terminate).
- Built a tamper-evident, hash-chained audit log with a CLI integrity-verification command for compliance-sensitive use; stood up full CI/CD from scratch (CodeQL, secret scanning, typed async internals via FastAPI and Pydantic v2, Docker image), staying backend-agnostic across cloud and local providers (OpenAI, Anthropic, Ollama).

Red_Co-Author | Python, Ollama, Streamlit, scikit-learn, Laminar

Author

- Engineered a fully-local Python red-team and defensive monitor stack for the Co-Authored Jailbreak (CoJP) LLM vulnerability, automating a two-armed attack pipeline (direct prompt vs CoJP-framed prompt) across multiple local target models orchestrated through Ollama and scoring both responses with a separate judge model on the HarDBench 1-to-5 harmfulness rubric.
- Trained a defensive monitor on the resulting attack dataset using local nomic-embed-text embeddings plus a scikit-learn logistic regression classifier; evaluated for generalization on held-out AdvBench (520 prompts) and HarDBench (400 prompts) splits and benchmarked head-to-head against a zero-shot llama3 baseline. Surfaced the full pipeline through a Streamlit dashboard with ASR-by-target heatmaps and ROC curves.

Curio (urban-toolkit/curio) | *Python, Flask, Socket.IO, React, TypeScript**Credited Contributor*

- Designed and built a real-time, multi-device collaboration layer for Curio, a dataflow framework for urban visual analytics, that lets multiple users edit the same workflow together over a shared network; presented the working feature to the project owner and core collaborators.
- Implemented the sync model on Socket.IO: live presence with per-user color coding, per-node soft locks with "Editing: name" badges, accept/decline code-change proposals with first-accept-wins conflict resolution, and execution-output propagation across collaborators' downstream nodes.
- Maintainers adopted the prototype into a **collaboration-revised** branch; the event vocabulary and proposal/lock model from the prototype ship in the released feature, with credit by name in the project's official documentation at [urban-toolkit/curio/docs/COLLABORATION.md](#).

PROJECTS**Hunter** | *Python, Textual, Ollama, httpx, SQLite, pytest**In Progress*

- Building a local-first, evidence-gated AI assistant for authorized penetration testing that runs a staged recon-to-report workflow fully offline, treating every model claim as a hypothesis that must be cross-checked against raw tool output before it can enter a report.
- Engineered a verification layer that extracts hard tokens (ports, versions, CVEs) from model output and cross-checks them against observed evidence, discarding and logging hallucinations; enforced a hard scope fence beneath a bounded autonomy loop (capped iterations, wall-clock, and actions) so no probe can touch an out-of-scope target.

Mini Tor Anonymity Network | *Python, TCP Sockets, Curve25519, AES-256-GCM, HKDF, SHA-256*

- Implemented Tor's three-hop onion routing protocol from scratch as six independent processes (directory, guard/middle/exit relays, client, destination) communicating over real TCP, with telescoping EXTEND cells so the client only ever connects to the guard while middle and exit are reached through the growing circuit.
- Built the entire cryptographic stack in pure Python with no third-party crypto libraries: SHA-256 (FIPS 180-4), HMAC, HKDF (RFC 5869), X25519 Diffie-Hellman via Montgomery ladder (RFC 7748), AES-256 (FIPS 197), and AES-256-GCM (NIST SP 800-38D), each verified against published NIST and IETF test vectors.

DiffFence | *Python, OSV.dev, Ollama*

- Built a local-first Python CLI that acts as a diff-aware security gate, scanning code changes for security vulnerabilities and production risks before a push or merge.
- Implemented baseline-on-first-run logic so subsequent scans surface only new findings, eliminating repeated noise and keeping developer focus on incremental risk introduced by each change.

KeyTracker | *Next.js, MongoDB, Vercel*

- Built and deployed a barcode-based key checkout system for UIC Technology Solutions, enabling staff to check keys in and out with real-time availability tracking.
- Implemented audit logging, conflict detection (409 responses for double-checkouts), and mismatch flagging to enforce access control integrity and maintain a reliable activity trail.

EDUCATION**University of Illinois Chicago (UIC)**

Chicago, IL

*Master of Science in Computer Science**December 2026*

- Relevant Coursework: Network Security, Cryptography, Computer System Security, Wearable Technology, Data Visualization & Analytics

University of Illinois Chicago (UIC)

Chicago, IL

*Bachelor of Science in Computer Science – Software Engineering Concentration; Minor in Mathematics**May 2025*

- Dean's List: College of Engineering – Fall 2023 & Fall 2024
- Relevant Coursework: Building Secure Computer Systems, Artificial Intelligence, Database Systems, Software Engineering 1 & 2, UI/UX, Software Design, Statistical Methods, Data Structures

CERTIFICATIONS**Foundations of Cybersecurity** – Google (Coursera)

Oct 2024

Introduction to Networking – NVIDIA (Coursera)

Oct 2024